Sharon Kodali

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Education

University of California – San Diego San Diego, CA

B.S. Data Science, Minor in Cognitive Science (Machine Learning), GPA: 4.00 *Expected 2028*

Relevant Courses: Data Structures & Algorithms, Systems Programming, Linear Algebra, Probability & Statistics, Discrete Mathematics

Experience

Undergraduate Research Intern Nov 2026 – Present

San Diego Supercomputer Center (SDSC) *La Jolla, CA*

- Porting a C++ fusion plasma simulation to JAX with Dr. Igor Sfiligoi for portable GPU execution (XLA/`jit`) and automatic differentiation (`grad`) for reactor-design optimization
- Building the port's regression suite: reading C++ routines to extract inputs, outputs, invariants and edge cases, then writing parametrized pytest tests against C++ reference outputs with `rtol`/`atol` tolerances
- Validating AI-generated and hand-written JAX kernels for numerical equivalence with the C++ originals

Software Development Intern (iOS) Jul 2026 – Present

Rydr *Remote*

- Cut API response time 62% (800ms to 300ms) by implementing local caching for a ride-sharing app
- Reduced crash rate 40% by debugging a memory leak; shipped profile editing and ride history via 25+ code-reviewed PRs
- Caught 12 bugs pre-release by writing 30+ unit and integration tests for the ride-booking payment flow (85% coverage)
- Resolved 10+ account-creation and 8 authentication bugs across the MySQL backend and REST APIs, including 3 security vulnerabilities that exposed other users' accounts
- Fixed 15+ data-integrity bugs in driver and rider records, including reward-point sync errors and form-validation gaps

Data & Machine Learning Analyst May 2026 – Aug 2026

Sprout Media *Remote*

- Built an automated Python pipeline aggregating 50K+ records/week from 3 APIs and 4 CSV sources at 99%+ success for 10 weeks
- Cut pipeline runtime 40% (60 to 36 min) by refactoring batch API calls into async requests; cut null and duplicate records 25% with validation checks
- Delivered 2 weekly-refreshed dashboards adopted by 5 stakeholders across 3 client accounts; surfaced a 15% underperforming campaign segment that drove a client budget reallocation
- Engineered features for the team's predictive models using LLM-based parsing of raw text data

Project Manager & Developer Nov 2024 – Jun 2025

Open Coding Society *San Diego, CA*

- Owned a Java (Spring Boot) and MySQL platform with REST APIs on AWS, used by 500+ students at 5 high schools, from data model to UI
- Delivered a 7-feature roadmap as product manager for a 30-person, 7-sub-team engineering org, prioritizing the backlog on user needs and capacity
- Cut bug turnaround 60% (5 to 2 days) by leading sprint planning, demos and code reviews (40+ PRs)
- Presented the roadmap and outcomes to the Poway Unified School District Board and District STEM Expo

Lead Research Intern Jun 2024 – May 2025

UCSD Institute of Engineering in Medicine *La Jolla, CA*

- Raised lab throughput 5x (3 hours to 20 minutes per dataset) by leading a 6-person team to build a Python/MATLAB feature-extraction pipeline for 500+ microscopy images
- Translated open-ended research questions into technical requirements as liaison between engineers and medical researchers
- Documented methods and results that contributed to published Biomedical Engineering Society (BMES) abstracts

Projects

Distributed Key-Value Store | *Go, Raft, Docker* Code

- Built a 5-node fault-tolerant Raft cluster in Go, achieving zero data loss and leader re-election in under 4s after killing the leader mid-write
- Sustained 3,200 writes/sec at 500 concurrent clients and sub-3ms p99 reads; wrote 100+ tests (89% coverage) under Go's race detector

- Fixed 2 distributed-systems bugs (an IP/hostname mismatch and a DNS startup race) that surfaced only under real network conditions; containerized the cluster with Docker
- Found and fixed 2 more bugs through new tests: a nil-snapshot restore panic and a leaked TCP port on failed startup

TritonETA – Real-Time Shuttle Tracker & ML Predictor | *Python, SQL, DuckDB, dbt, GitHub Actions* Code
- Built an automated ingestion pipeline polling a live GTFS-RT GPS feed every 15–30s via GitHub Actions, collecting 180K GPS pings from 28 shuttles on 6 routes unattended
- Modeled the data in 16 version-controlled dbt models with 28 data tests (including a train/test leakage test) and documented lineage
- Engineered a geospatial matching algorithm mapping GPS pings to 61 scheduled stops, deriving an arrival-delay signal missing from the source feed (30,946 stop visits matched, 63% on time)
- Produced a 27,166-row labeled training set with a time-based train/test split for the ETA model

xView2 Building Damage Segmentation | *Python, PyTorch, U-Net* Code
- Built a PyTorch U-Net that segments per-building disaster damage from paired pre/post satellite images (xBD), reaching 0.475 damage macro F1 vs. 0.185 for a baseline given the true building outlines
- Showed pixel accuracy was misleading (92.8% of pixels are background) and adopted building-level macro F1, using a class-weighted loss to recover minority damage classes
- Wrote 42 pytest tests on synthetic data with CPU-only CI; the tests caught a flip-augmentation bug in the training code

ResumeTailor – AI Resume Tailoring Platform | *Python, FastAPI, Next.js, TypeScript, OpenAI API* Code | Demo
- Built a full-stack LLM app that tailors resumes to job postings by rewriting LaTeX bullets in place via offset splicing, so document structure never breaks
- Added a relevance-scoring metric to check model output before trusting it; covered the backend with 96 pytest tests and CI
- Cut tailoring time 83% (30 to under 5 min) and response latency 45% with caching and async batching

VibeCanvas – ML Generative Art from Spotify Audio | *Python, pandas, scikit-learn, JavaScript* Code | Demo
- Cleaned and explored 28,351 Spotify tracks; PCA showed 9 components explain 95% of audio-feature variance
- Clustered tracks into 4 natural audio groups and classified 6 genres with a Random Forest at 54% accuracy (logistic regression 46%, chance 16.7%), using stratified 5-fold CV
- Rendered each song's audio fingerprint as generative art in a live browser demo

BlinkGuard – Real-Time Driver Drowsiness Detection | *TypeScript, Next.js, MediaPipe, Python, Fetch.ai* Code | D
- Co-built a browser app that tracks eye and mouth landmarks (MediaPipe Face Mesh) to compute eye- and mouth-aspect ratios and detect long blinks, closed eyes and yawns in real time
- Personalized alert thresholds with per-driver calibration and a Fetch.ai multi-agent service that escalates alerts and scores trips

WattWise SD – Utility Bill Explainer | *TypeScript, Next.js, Gemini API* Code | Demo
- Built at Hack for Humanity @ UCSD: explains San Diego electricity bills in plain language and 5 languages and finds the most expensive usage hours, with tests and CI

UCSD Deadlines – Browser Extension | *TypeScript, React, WXT* Code
- Built a browser extension that pulls Canvas assignments through the Planner API using the existing session (no stored credentials) into Today, List and Week views

Language Model & Autograd Engine from Scratch | *Python, NumPy* 2026
- Reduced validation loss from 10.5 to 1.45 by building a language model with no ML frameworks
- Implemented forward and backward passes in NumPy by hand-deriving backpropagation via the chain rule

B+ Tree Index | *C++, Data Structures* 2026
- Implemented a database index in C++ with search, insertion, and node splits that keep the tree balanced under arbitrary insertion order

Technical Skills

Languages: Python, Go, C/C++, Java, SQL, JavaScript, TypeScript, MATLAB, Bash
ML & Data: PyTorch, JAX, scikit-learn, pandas, NumPy, MediaPipe, LLM APIs (OpenAI, Gemini), dbt, DuckDB, PostgreSQL, MySQL
Systems: Distributed Systems, Raft Consensus, Concurrency, Networking, Docker, REST APIs, AWS
Web: FastAPI, Next.js, React, Spring Boot, Flask
Tools: Git, CI/CD (GitHub Actions), pytest, Testing & Debugging, Agile, Claude Code, Cursor